

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

ANSWER:

Problem Statement

A university has been allocated the IPv4 address block **192.168.20.0/24** for its internal network. The network administrator needs to divide this network into three subnets using fixed subnet masks for different laboratories. The subnet requirements are:

- **Lab A:** /26
- **Lab B:** /27
- **Lab C:** /28

Determine the following for each laboratory:

1. Network Address

2. Valid Host IP Address Range
3. Broadcast Address
4. Total Number of Usable Host IP Addresses

Given Data

Parameter	Value
Base Network Address	192.168.20.0
Prefix Length	/24
Lab A Subnet	/26
Lab B Subnet	/27
Lab C Subnet	/28

Objective

To divide the given **192.168.20.0/24** network into three subnets using the specified subnet masks and determine the network address, host address range, broadcast address, and usable host addresses for each subnet.

Method / Formula Used

Formula 1: Number of Usable Hosts

Usable Hosts = $2^{(32 - \text{Prefix Length})} - 2$
Usable Hosts = $2^{(32 - \text{Prefix Length})} - 2$

Formula 2: Block Size

Block Size = $256 - \text{Last Octet of Subnet Mask}$
Block Size = $256 - \text{Last Octet of Subnet Mask}$

Solution

Step 1: Subnet for Lab A (/26)

Subnet Mask

255.255.255.192

Block Size

$256 - 192 = 64$

Hence, subnet intervals are

0, 64, 128, 192

The first subnet starts from **192.168.20.0**

Network Address

192.168.20.0

Broadcast Address

192.168.20.63

Valid Host Range

192.168.20.1

to

192.168.20.62

Number of Usable Hosts

$2^{(32-26)} - 2 = 2^6 - 2 = 64 - 2 = 62$

$2^6 - 2 = 64 - 2 = 62$

$64 - 2 = 62$

Therefore,

Usable Hosts = 62

Step 2: Subnet for Lab B (/27)

The next available subnet begins at **192.168.20.64**

Subnet Mask

255.255.255.224

Block Size

$256 - 224 = 32$

Network Address

192.168.20.64

Broadcast Address

192.168.20.95

Valid Host Range

192.168.20.65

to

192.168.20.94

Number of Usable Hosts

$$2(32-27)-22^{\{(32-27)\}}-22(32-27)-2$$

$$25-22^5-225-2$$

$$32-2=30 \quad 32-2=30 \quad 32-2=30$$

Therefore,

Usable Hosts = 30

Step 3: Subnet for Lab C (/28)

The next available subnet begins at **192.168.20.96**

Subnet Mask

$$255.255.255.240$$

Block Size

$$256-240=16 \quad 256-240=16 \quad 256-240=16$$

Network Address

$$192.168.20.96$$

Broadcast Address

$$192.168.20.111$$

Valid Host Range

$$192.168.20.97$$

to

$$192.168.20.110$$

Number of Usable Hosts

$$2(32-28)-22^{\{(32-28)\}}-22(32-28)-2$$

$$24-22^4-224-2$$

$$16-2=14 \quad 16-2=14 \quad 16-2=14$$

Therefore,

Usable Hosts = 14

2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet

ANSWER:

Problem Statement

A network administrator has configured two systems with the following network settings:

- **System 1:** IP Address = **192.168.1.130**, Subnet Mask = **255.255.255.192 (/26)**
- **System 2:** IP Address = **192.168.1.190**, Subnet Mask = **255.255.255.192 (/26)**

Using the given configuration, apply subnetting techniques to:

1. Determine the subnet range for each system.
2. Find the network address and broadcast address.
3. Identify the valid host IP address range.
4. Verify whether each IP address belongs to its respective subnet.
5. Calculate the total number of usable host addresses available in the subnet.

Given Data

Parameter	System 1	System 2
IP Address	192.168.1.130	192.168.1.190
Subnet Mask	255.255.255.192	255.255.255.192
Prefix Length	/26	/26

Objective

To determine the subnet information for both systems by calculating the network address, subnet range, broadcast address, valid host range, and the number of usable host addresses using the given subnet mask.

Method / Formula Used

Formula 1: Block Size

Block Size = $256 - \text{Last Octet of Subnet Mask}$
 $\text{Block Size} = 256 - \text{Last Octet of Subnet Mask}$

Formula 2: Number of Usable Hosts

Usable Hosts = $2^{(32 - \text{Prefix Length})} - 2$
 $\text{Usable Hosts} = 2^{(32 - \text{Prefix Length})} - 2$

Solution

Step 1: Determine the Block Size

Subnet Mask = **255.255.255.192**

Last octet = **192**

Block Size = $256 - 192 = 64$
 $\text{Block Size} = 256 - 192 = 64$

Hence, the subnet ranges are:

Subnet	Address Range
1	192.168.1.0 – 192.168.1.63
2	192.168.1.64 – 192.168.1.127
3	192.168.1.128 – 192.168.1.191
4	192.168.1.192 – 192.168.1.255

Both IP addresses (**192.168.1.130** and **192.168.1.190**) fall within the **third subnet (192.168.1.128 – 192.168.1.191)**.

Step 2: Determine the Network Address

Network Address:

192.168.1.128

Step 3: Determine the Broadcast Address

Broadcast Address:

192.168.1.191

Step 4: Determine the Valid Host IP Range

The first usable host address is:

192.168.1.129

The last usable host address is:

192.168.1.190

Therefore,

Valid Host Range:

192.168.1.129 – 192.168.1.190

Step 5: Verify the Given IP Addresses

System 1

Given IP Address:

192.168.1.130

Since **192.168.1.130** lies between **192.168.1.129** and **192.168.1.190**, it is a **valid host address** in this subnet.

System 2

Given IP Address:

192.168.1.190

Since **192.168.1.190** lies between **192.168.1.129** and **192.168.1.190**, it is also a **valid host address** in the same subnet.

Step 6: Calculate the Number of Usable Hosts

For a **/26** subnet,

Usable Hosts = $2^{(32-26)} - 2$ = $2^{\{(32-26)\}}$ -

$$2^6 \text{ Usable Hosts} = 2^{(32-26)} - 2 = 2^6 - 2 = 2^6 - 2 = 64 - 2 = 64 - 2 = 62 = 62$$

Therefore,

Total Usable Host Addresses = 62

3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin →

/28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department. Finally, identify the remaining unused IP address range within the given network block.

ANSWER:

Problem Statement

An organization has been allocated the IPv4 address block **172.16.0.0/16** for its internal network. The network administrator has decided to allocate separate subnets for different departments using the following subnet masks:

- **HR Department** → /25
- **Sales Department** → /26
- **IT Department** → /27
- **Admin Department** → /28

Using the given subnet masks, determine:

1. The network address for each department.
2. The valid host IP address range for each subnet.
3. The broadcast address for each subnet.
4. The total number of usable host addresses available for each department.

5. The remaining unused IP address range within the given network block.

Given Data

Parameter	Value
Base Network Address	172.16.0.0
Prefix Length	/16
HR Department	/25
Sales Department	/26
IT Department	/27
Admin Department	/28

Objective

To apply subnetting techniques for dividing the given **172.16.0.0/16** network into multiple departmental subnets and determine the network address, valid host range, broadcast address, usable host addresses, and remaining unused address range.

Method / Formula Used

Formula 1: Block Size

Block Size=256–Last Octet of Subnet Mask
 $\text{Block Size} = 256 - \text{Last Octet of Subnet Mask}$

Formula 2: Number of Usable Hosts

Usable Hosts=2(32–Prefix Length)–2
 $\text{Usable Hosts} = 2^{(32 - \text{Prefix Length})} - 2$

Solution

Step 1: Allocate Subnet for HR Department (/25)

Subnet Mask

255.255.255.128

Block Size

256–128=128
 $256 - 128 = 128$

Network Address

172.16.0.0

Broadcast Address

172.16.0.127

Valid Host Range

172.16.0.1 – 172.16.0.126

Number of Usable Hosts

$2^{(32-25)} - 2 = 2^7 - 2 = 128 - 2 = 126$

$2^7 - 2 = 128 - 2 = 126$

$128 - 2 = 126$

Therefore,

Usable Hosts = 126

Step 2: Allocate Subnet for Sales Department (/26)

The next available subnet starts at **172.16.0.128**.

Subnet Mask

255.255.255.192

Block Size

$256 - 192 = 64$

Network Address

172.16.0.128

Broadcast Address

172.16.0.191

Valid Host Range

172.16.0.129 – 172.16.0.190

Number of Usable Hosts

$2^{(32-26)} - 2 = 2^6 - 2 = 64 - 2 = 62$

$2^6 - 2 = 64 - 2 = 62$

$64 - 2 = 62$

Therefore,

Usable Hosts = 62

Step 3: Allocate Subnet for IT Department (/27)

The next available subnet starts at **172.16.0.192**.

Subnet Mask

255.255.255.224

Block Size

$256 - 224 = 32$

Network Address

172.16.0.192

Broadcast Address

172.16.0.223

Valid Host Range

172.16.0.193 – 172.16.0.222

Number of Usable Hosts

$2^{(32-27)} - 2 = 32 - 2 = 30$

$2^5 - 2 = 32 - 2 = 30$

Therefore,

Usable Hosts = 30

Step 4: Allocate Subnet for Admin Department (/28)

The next available subnet starts at **172.16.0.224**.

Subnet Mask

255.255.255.240

Block Size

$256 - 240 = 16$

Network Address

172.16.0.224

Broadcast Address

172.16.0.239

Valid Host Range

172.16.0.225 – 172.16.0.238

Number of Usable Hosts

$2^{(32-28)} - 2 = 16 - 2 = 14$

$2^4 - 2 = 16 - 2 = 14$

$$16-2=14 \quad 16-2=14 \quad 16-2=14$$

Therefore,

Usable Hosts = 14

Step 5: Determine the Remaining Unused IP Address Range

After allocating subnets for all four departments, the remaining address space begins from:

172.16.0.240

and extends up to the last address in the allocated /16 network:

172.16.255.255

Therefore,

Remaining Unused IP Address Range:

Department	Network Address	Valid Host Range	Broadcast Address	Usable Hosts
HR	172.16.0.0/25	172.16.0.1 – 172.16.0.126	172.16.0.127	126
Sales	172.16.0.128/26	172.16.0.129 – 172.16.0.191	172.16.0.192	62
IT	172.16.0.192/27	172.16.0.193 – 172.16.0.222	172.16.0.223	30
Admin	172.16.0.224/28	172.16.0.225 – 172.16.0.238	172.16.0.239	14

Department	Network Address	Valid Host Range	Broadcast Address	Usable Hosts
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8

172.16.0.240 – 172.16.255.255

Result

Remaining Unused IP Address Range

172.16.0.240 – 172.16.255.255

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

ANSWER:

Problem Statement

A company uses the IPv4 network **172.16.10.0/24** for its internal departments and has implemented subnetting using a **/27** subnet mask.

Using the given subnet mask, determine:

1. The total number of subnets created.
2. The number of usable host addresses available in each subnet.
3. The subnet to which the IP address **172.16.10.78** belongs.
4. The network address and broadcast address for that subnet.
5. Whether the IP address **172.16.10.95** falls within the same subnet based

on the given configuration.

Given Data

Parameter	Value
Base Network Address	172.16.10.0
Prefix Length	/24
Subnet Mask	/27 (255.255.255.224)
Given IP Address 1	172.16.10.78
Given IP Address 2	172.16.10.95

Objective

To apply subnetting concepts to determine the number of subnets, usable host addresses per subnet, identify the subnet containing the given IP address, determine its network and broadcast addresses, and verify whether another IP address belongs to the same subnet.

Method / Formula Used

Formula 1: Number of Subnets

Number of Subnets = $2^{(\text{New Prefix} - \text{Original Prefix})}$
 $2^{(\text{New Prefix} - \text{Original Prefix})}$
Number of Subnets = $2^{(\text{New Prefix} - \text{Original Prefix})}$

Formula 2: Number of Usable Hosts

Usable Hosts = $2^{(32 - \text{Prefix Length})} - 2$
Usable Hosts = $2^{(32 - \text{Prefix Length})} - 2$

Formula 3: Block Size

Block Size = $256 - \text{Last Octet of Subnet Mask}$
Block Size = $256 - \text{Last Octet of Subnet Mask}$

Solution

Step 1: Calculate the Number of Subnets

Original Prefix = /24

New Prefix = /27

Borrowed Bits

$$27-24=3 \quad 2^3-2=3$$

Number of Subnets

$$2^3=8 \quad 2^3-2=6$$

Therefore,

Total Number of Subnets = 8

Step 2: Calculate the Number of Usable Hosts

$$2^{(32-27)}-2=2^5-2=30 \quad 2^5-2=30$$

$$32-2=30 \quad 32-2=30$$

Therefore,

Usable Host Addresses per Subnet = 30

Step 3: Determine the Subnet of IP Address 172.16.10.78

Block Size

$$256-224=32 \quad 256-224=32$$

The subnet ranges are:

Subnet	Address Range
1	172.16.10.0 – 172.16.10.31
2	172.16.10.32 – 172.16.10.63
3	172.16.10.64 – 172.16.10.95
4	172.16.10.96 – 172.16.10.127
5	172.16.10.128 – 172.16.10.159
6	172.16.10.160 – 172.16.10.191
7	172.16.10.192 – 172.16.10.223
8	172.16.10.224 – 172.16.10.255

The IP address **172.16.10.78** falls within the subnet **172.16.10.64 – 172.16.10.95**.

Therefore,

Network Address = 172.16.10.64

Step 4: Determine the Broadcast Address

The last address of the subnet is the broadcast address.

Broadcast Address = 172.16.10.95

Step 5: Determine the Valid Host Range

First Usable Host Address

172.16.10.65

Last Usable Host Address

172.16.10.94

Therefore,

Valid Host Range = 172.16.10.65 – 172.16.10.94

Step 6: Verify Whether 172.16.10.95 Belongs to the Same Subnet

The IP address **172.16.10.95** is the **broadcast address** of the subnet.

Although it belongs to the same subnet, it **cannot be assigned to any host** because broadcast addresses are reserved for sending data to all devices within the subnet.

Therefore,

172.16.10.95 belongs to the same subnet but is not a valid host IP address.

Result

Parameter	Value
Network Address	172.16.10.64
Broadcast Address	172.16.10.95
Valid Host Range	172.16.10.65 – 172.16.10.94
Number of Subnets	8
Usable Hosts per Subnet	30
Subnet of 172.16.10.78	172.16.10.64/27
Does 172.16.10.95 belong to	Yes, but it is the broadcast address and

Parameter

the same subnet?

Value

cannot be used as a host address.

5.A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

ANSWER:

Problem Statement

A network uses the IPv6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to:

1. Compress the given IPv6 address into its shortest notation.
2. Expand the compressed address back to its full notation.
3. Determine the network prefix using a **/64** prefix length.
4. Calculate the total number of possible host addresses available within the network.

Given Data

Parameter	Value
IPv6 Address	2001:0db8:0000:0000:0000:ff00:0042:8329
Prefix Length	/64

Objective

To apply IPv6 addressing rules for compressing and expanding the given IPv6 address, determine the network prefix using a **/64** prefix length, and calculate the total number of possible host addresses available in the network.

Method / Formula Used

Rule 1: IPv6 Compression

- Remove leading zeros in each hexadecimal block.

- Replace one consecutive sequence of zero blocks with :: (only once).

Rule 2: IPv6 Expansion

- Replace :: with the required number of **0000** blocks.
- Add leading zeros so that each block contains four hexadecimal digits.

Formula: Number of Host Addresses

Host Bits = 128 – Prefix Length

Host Bits = 128 – Prefix Length

Total Host Addresses = $2^{\text{Host Bits}}$

Solution

Step 1: Compress the IPv6 Address

Given IPv6 Address:

2001:0db8:0000:0000:0000:ff00:0042:8329

Remove Leading Zeros

2001:db8:0:0:0:ff00:42:8329

Compress Consecutive Zero Blocks

Three consecutive **0** blocks are replaced with ::.

Compressed IPv6 Address:

2001:db8::ff00:42:8329

Step 2: Expand the IPv6 Address

Compressed Address:

2001:db8::ff00:42:8329

Replace :: with three **0000** blocks and restore leading zeros.

Expanded IPv6 Address:

2001:0db8:0000:0000:0000:ff00:0042:8329

Step 3: Determine the Network Prefix

The prefix length is /64, which means the **first 64 bits** (first four hexadecimal blocks) represent the network portion.

Network Prefix:

2001:0db8:0000:0000::/64

or

2001:db8:0:0::/64

Step 4: Calculate the Total Number of Host Addresses

Host Bits

$$128 - 64 = 64 \quad 128 - 64 = 64$$

Total Possible Host Addresses

$$2^{64} = 18,446,744,073,709,551,616$$

$$18,446,744,073,709,551,616 = 18,446,744,073,709,551,616$$

Therefore,

Total Possible Host Addresses = 18,446,744,073,709,551,616 (approximately 18.4 quintillion).

Result

Parameter	Result
Original IPv6 Address	2001:0db8:0000:0000:0000:ff00:0042:8329
Compressed IPv6 Address	2001:db8::ff00:42:8329
Expanded IPv6 Address	2001:0db8:0000:0000:0000:ff00:0042:8329
Network Prefix (/64)	2001:0db8:0000:0000::/64
Host Bits	64
Total Possible Host Addresses	$2^{64} = 18,446,744,073,709,551,616$

Interpretation

The given IPv6 address has been successfully compressed and expanded according to IPv6 addressing rules. Using a /64 prefix length, the first 64 bits represent the network portion, while the remaining 64 bits are used for host addressing. A /64 IPv6 network supports 2^{64} (**18,446,744,073,709,551,616**) possible host addresses, making it the standard subnet size for IPv6 networks

and providing an extremely large address space for modern networking applications.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

